Relational Graph-Guided Selective State-Space Model for Hyperspectral Image Classification

Muhammad Ahmad, Hamzah Luqman, and Manuel Mazzara

***Abstract*—Hyperspectral image (HSI) classification remains challenging under extreme label scarcity and spatially disjoint supervision, where models must generalize across heterogeneous materials while producing spatially consistent maps at scale. This article proposes RGMamba (RGM), a relational graph-guided selective state-space model (SSSM) for HSI classification in scarce-label, disjoint settings. RGM constructs physically grounded relational neighborhoods by jointly enforcing spatial proximity and spectral affinity, thereby promoting materially coherent context. Neighborhood-derived distributional statistics are fused with spectral signatures to enrich per-pixel representations while preserving pixel-to-map correspondence. These graph-derived descriptors are directly interpretable in terms of spatial–spectral neighborhood consistency and spectral homogeneity; however, the subsequent dual-branch encoder and SSSM model are learned transformations and do not share the same level of transparency. A pixel-aligned dual-branch encoder integrates local spatial aggregation and bandwise spectral mixing without patch pooling, and an SSSM captures long-range dependencies with linear-time sequence complexity. While the SSSM itself scales linearly with sequence length, the overall framework incurs additional computational overhead from relational neighborhood construction, graph-statistics extraction, and pixel-aligned token processing. Experiments on six benchmark datasets show that RGM consistently outperforms representative Transformer and state-space baselines under scarce supervision. Under a stringent spatially disjoint protocol with 0.05% labeled samples, RGM achieves over 96% overall accuracy.**

***Index Terms*—Hyperspectral image (HSI) classification, Mamba, relational graphs, selective state-space models.**

I. Introduction

HYPERSPECTRAL Image (HSI) classification leverages hundreds of contiguous spectral bands to discriminate materials at the pixel level [1]. While the rich spectral resolution enables fine-grained recognition [2], it also introduces challenges such as high dimensionality [3], strong spectral redundancy [4], intricate spectral–spatial correlations [5], and scarce, imbalanced labels [6]. Robust solutions must therefore fuse spectral affinity with spatial continuity, preserve interpretable pixel-level relational descriptors, and model long-range dependencies efficiently and reliably under limited supervision [7].

Conventional machine learning models, such as support vector machines (SVMs), k-nearest neighbors (kNNs), and shallow artificial neural networks (ANNs), remain competitive in low-data regimes but typically operate on hand-crafted or pixel-wise features. Moreover, these models ignore spatial context and require delicate preprocessing to mitigate the "curse of dimensionality" [8]. In contrast, traditional deep learning improved spectral–spatial fusion yet exhibits several structural limitations [9]. 2-D/3-D convolutional neural networks (CNNs) encode locality but require deep stacks or large kernels to capture long-range interactions [10], which results in an increasing number of parameters and a risk of oversmoothing [11]. Sequence models such as recurrent neural networks (RNNs), long short-term memory (LSTM), and gated recurrent units (GRUs) capture spectral ordering but struggle with long contexts [12], exhibit training instability, and often decouple spectral from spatial reasoning [13]. Graph-based methods introduce a relational inductive bias, but they often rely on fragile heuristics, such as kNN in the image plane [14], that conflate geometric proximity with true spectral similarity. In addition, dense graphs can lead to scalability issues, and message passing may blur boundaries or oversmooth features [15], [16].

Recent attention models, such as Transformers and graph Transformers, explicitly model long-range dependencies [17]; however, they entail quadratic complexity in terms of token or node count, requiring extensive data and careful regularization [18]. Moreover, common tokenization strategies break pixel-token alignment (e.g., large patches), which complicates analysis and interpretability [19]. Recently, state-space models (SSMs), such as Mamba, offer linear-time sequence modeling and strong long-context retention [20]. However, SSMs depend on token quality and often lack an explicit relational bias [21], [22]. Existing graph-augmented SSM pipelines either inherit the graph construction issues and do not disentangle spatial proximity from spectral affinity [23], which limits robustness and interpretability [24].

To overcome these limitations, this work proposes RGMamba (RGM), a relational graph-guided selective SSM (SSSM) for HSI classification under scarce-label and spatially disjoint settings. The proposed method is built on a principled relational modeling perspective, where spatial proximity and

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spectral affinity are jointly enforced to construct materially coherent neighborhoods. These neighborhoods are summarized via distributional statistics and integrated into pixel-level representations without spatial aggregation, preserving pixel-to-map correspondence. A pixel-aligned dual-branch encoder then produces structured tokens that retain both local spatial context and spectral relationships. Finally, an SSSM captures long-range dependencies with linear sequence complexity, enabling efficient long-context sequence modeling. Although the SSSM scales linearly with token length, the complete framework incurs additional computational costs for relational neighborhood construction, graph-based feature extraction, and pixel-aligned token generation. In a nutshell, this work makes the following contributions.

- C1 *Constrained Relational Graph Formulation:* We introduce an intersection-based relational graph, where spatial proximity and spectral affinity are jointly enforced through $E_{\text{spatial}} \cap E_{\text{spectral}}$. This formulation imposes a structural constraint that removes heterophilous connections and yields materially coherent neighborhoods grounded in spatial–spectral consistency.
- C2 *Pixel-Preserving Relational Representation:* We propose graph-derived distributional descriptors computed directly at the pixel level, without spatial aggregation. This design preserves strict pixel-to-map correspondence while encoding material homogeneity, improving robustness under scarce supervision.
- C3 *Graph-Guided Tokenization for Sequence Modeling:* We introduce a pixel-aligned dual-branch encoder that integrates spatial context and spectral mixing while maintaining one-to-one pixel-token alignment. This enables structured and physically grounded token representations suitable for long-context modeling.
- C4 *Efficient Long-Context Modeling via SSSM:* We integrate an SSSM with input-conditioned dynamics, enabling efficient long-range dependency modeling with linear complexity $\mathcal{O}(L\,d_s)$, providing a linear-complexity sequence-modeling alternative to quadratic attention mechanisms while acknowledging the additional computational overhead introduced by relational preprocessing and token construction.

II. Literature Review

Conventional machine learning models for HSI classification predominantly rely on per-pixel spectra and supervised classifiers such as SVM [25], kNN [26], and ANN [27]. These methods remain competitive in low-data regimes due to strong inductive biases and modest capacity [28]. However, their reliance on isolated spectral signatures limits their ability to explicitly model spatial continuity, which is often predictive in land-cover mapping [29]. Moreover, dimensionality reduction or band selection is typically required to mitigate the curse of dimensionality [30], and performance may degrade under illumination variability, atmospheric residuals, or sensor shifts without additional normalization or adaptation mechanisms [31]. Spatial context has been incorporated through morphological profiles, texture descriptors, and superpixel pooling [32], though such approaches remain partially hand-crafted, may increase engineering complexity [33], and can struggle in highly heterogeneous scenes [34].

CNNs have become a dominant paradigm for spectral–spatial modeling [35]. 2-D CNNs apply spatial filters over stacked spectral bands, whereas 3-D CNNs convolve jointly across (H, W, C) [36]. CNNs effectively encode local spatial structure; however, modeling broader contextual interactions typically requires deeper stacks, dilated convolutions, or enlarged receptive fields [10], which increase parameter counts and may introduce oversmoothing effects [37]. In scarce-label regimes, 3-D convolutions can become data-intensive and may entangle spectral and spatial cues in ways that complicate interpretability [38]. Hybrid architectures that decouple spectral and spatial streams alleviate some issues but often rely on hierarchical pyramids or deep stacks to approximate long-range dependencies [39].

Recurrent models, including RNNs, LSTMs, and GRUs, have been applied to model spectral ordering and extended to spatial reasoning via row/column or patchwise scans [40]. While these approaches provide sequential inductive bias, very long token sequences can introduce optimization challenges and memory limitations [41]. Practical implementations often rely on truncated contexts or auxiliary mechanisms to stabilize training [42]. Furthermore, spatial–spectral fusion in recurrent pipelines frequently requires additional preprocessing or restructuring steps [43], and gated recurrences may exhibit diminishing long-context retention when spectra are unfolded into long spatial token streams ($L = H\,W$) [44], [45].

Graph neural networks (GNNs) introduce relational inductive bias by operating on pixel- or superpixel-level graphs [46]. Graph construction plays a central role in their effectiveness [16]. Common strategies include kNN in the image plane or spectral space, weighted similarity graphs, or clustering-based approaches. While effective in many settings, such constructions may mix geometric proximity with spectral similarity in ways that depend on heuristic design choices [47]. Dense graphs can increase computational burden, and repeated message passing may smooth features across boundaries [48]. Beyond standard message-passing GNNs, recent work has explored stronger graph reasoning and structure modeling for HSI. For example, NSCKL [49] constructs similarity graphs through normalized spectral clustering with kernel-based learning in a semisupervised setting, providing a principled mechanism to propagate label information through spectrally coherent structures. GraphGST [50] further integrates graph structure into a Transformer framework by generating structure-aware representations, highlighting the potential of combining graph topology modeling with global dependency learning. These recent graph-based approaches have advanced structure-aware spectral–spatial learning; however, their evaluation settings often differ from the spatially disjoint protocol adopted in this work, making direct comparison dependent on the underlying experimental design. Related directions also include learned adjacency and graph-aware Transformer architectures [51], [52], which jointly optimize graph structure and feature propagation. Collectively, these advances demonstrate the effectiveness of graph-guided reasoning for spectral–spatial learning, while also motivating

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careful consideration of scalability, graph construction choices, and training complexity in large-scene settings.

Transformer-based models have advanced HSI classification by modeling global interactions through attention mechanisms [53]. Full attention entails quadratic complexity in token count ($\mathcal{O}(L^2)$) and substantial memory usage [20], motivating hierarchical tokenization, patch pooling, or sparsity priors [24], [54]. Sparse and linear-attention variants reduce computational cost but often trade expressivity or require problem-specific design choices [55], [56]. Graph-aware Transformer architectures further integrate relational structure into attention [52], improving structural modeling at the expense of increased architectural complexity.

Recent advances have also explored related challenges beyond supervised HSI classification. For example, S4DL [57] introduces shift-sensitive spatial–spectral disentangling to improve HSI domain adaptation under distribution shifts, while the multiteacher multiobjective meta-learning framework in [58] addresses zero-shot heterogeneous hyperspectral band selection through transferable meta-learning strategies. These studies highlight the growing importance of robust representation learning and generalization in hyperspectral analysis, complementing ongoing developments in spectral–spatial modeling and structure-aware learning.

SSMs, including recent Mamba-style architectures [20], provide linear-time sequence modeling and strong long-context retention. These models have been adapted for spectral–spatial analysis in remote sensing. However, their effectiveness depends heavily on token quality and representation structure [21]. When tokenization breaks pixel-to-map correspondence (e.g., via patch pooling), interpretability and boundary fidelity may be affected. Furthermore, graph-augmented SSM variants inherit the characteristics of their underlying graph construction mechanisms.

Collectively, prior work highlights the importance of integrating spatial continuity, spectral affinity, relational structure, and long-context modeling. The framework proposed in this study seeks to unify these elements through explicitly constrained relational neighborhoods based on spatial–spectral intersection, homogeneity-aware statistical enrichment at the pixel level, pixel-aligned dual-branch tokenization, and SSSM with linear sequence complexity, while preserving interpretability under spatially disjoint supervision.

III. Proposed Methodology

An HSI cube is denoted as $\mathcal{X} \in \mathbb{R}^{H\times W\times C}$, where H and W denote spatial dimensions and C is the number of spectral bands [59]. Each pixel at spatial location (h, w) is associated with a spectral vector $\mathbf{x}_{h,w} \in \mathbb{R}^C$, with elements $\mathbf{x}_{h,w}(c)$ representing reflectance at band c [60]. When labels are available, the corresponding ground-truth class for pixel (h, w) is denoted by $y_{h,w} \in \mathbb{Z}$ (e.g., vegetation type, mineral composition) [61]. The complete model is shown in Fig. 1 and Algorithm 1. The proposed framework is designed around a constrained relational modeling objective, where spatial proximity and spectral affinity are jointly enforced to capture materially coherent neighborhoods while preserving pixel-level semantics for downstream sequence modeling.

Algorithm 1 RGMamba

Require: HSI cube, labels, patch size, spatial threshold T_d, spectral threshold T_s
Ensure: Pixelwise posteriors

Extract overlapping patches of size $P \times P \times C$ with stride s
Compute per-band (μ_c, σ_c) on training patches only
for each patch **do**
  Create nodes $V = \{v_{h,w}\}$ for all pixels (h, w) in the patch
  Compute **spatial relations** from edges using Equation (3)
  Compute **spectral relations** from edges using Equation (5)
  for each node $v_{h,w} \in V$ **do**
    Retrieve neighborhood using Equation (8)
    Compute $F_{h,w} = [C, S_{\text{avg}}, S_{\max}, S_{\min}, S_{\text{var}}]$
    Concatenate $\mathbf{f}_{h,w} \leftarrow [\mathbf{x}_{h,w}, F_{h,w}] \in \mathbb{R}^{C+5}$
  end for
  Stack $\mathbf{f}_{h,w}$ over (h, w) to form enriched patch feature map
end for
For each $\mathbf{U}^{(n)}$, compute $\mathbf{T}^{(\text{sp})}$ and $\mathbf{T}^{(\text{spec})}$
Flatten to $L = P^2$ tokens and concatenate $\mathbf{E}^{(n)} \leftarrow \text{LayerNorm}([\mathbf{T}^{(\text{sp})}, \mathbf{T}^{(\text{spec})}])$
Process token sequence $\mathbf{E}^{(n)} = (\mathbf{e}_1, \ldots, \mathbf{e}_L)$ with state $\mathbf{s}_0 = \mathbf{0}$
for $t = 1$ to L **do**
  Compute input-conditioned parameters Δ_t and $\mathbf{B}_t$
  Update selective discretization $\overline{\mathbf{A}}_t, \overline{\mathbf{B}}_t$ and state $\mathbf{s}_t$
end for
Obtain global representation $\mathbf{h}_L$
Compute posteriors $\hat{\mathbf{y}} = \text{softmax}(\text{MLP}(\mathbf{h}_L))$
Minimize $\mathcal{L}$ with dropout regularization
return Posteriors probabilities $\mathbf{y}$

A. Patch Extraction

To capture both local spectral information and spatial context, the HSI cube is divided into overlapping 3-D patches [62]. Each patch is centered at spatial coordinates (α, β) and spans $P \times P$ pixels over the C spectral bands, forming a localized 3-D subcube [63]. This representation is crucial because spectral information alone often fails to discriminate between classes with similar signatures, whereas incorporating local spatial context improves robustness.

The total number of extracted patches is given by $m = (H - P + 1) \times (W - P + 1)$, assuming a stride of $s = 1$. More generally, patches are extracted with a stride $s \leq P$. Overlapping patches allow neighboring regions to share information, which is particularly important for land-cover mapping, where spatial continuity carries semantic meaning. The extent of overlap is quantified by the overlap ratio $r = 1 - s/P$, where $r = 0$ corresponds to nonoverlapping patches and $r > 0$ indicates overlap. Smaller stride values increase overlap and contextual redundancy, helping preserve fine-grained boundaries and subtle spectral differences across adjacent pixels.

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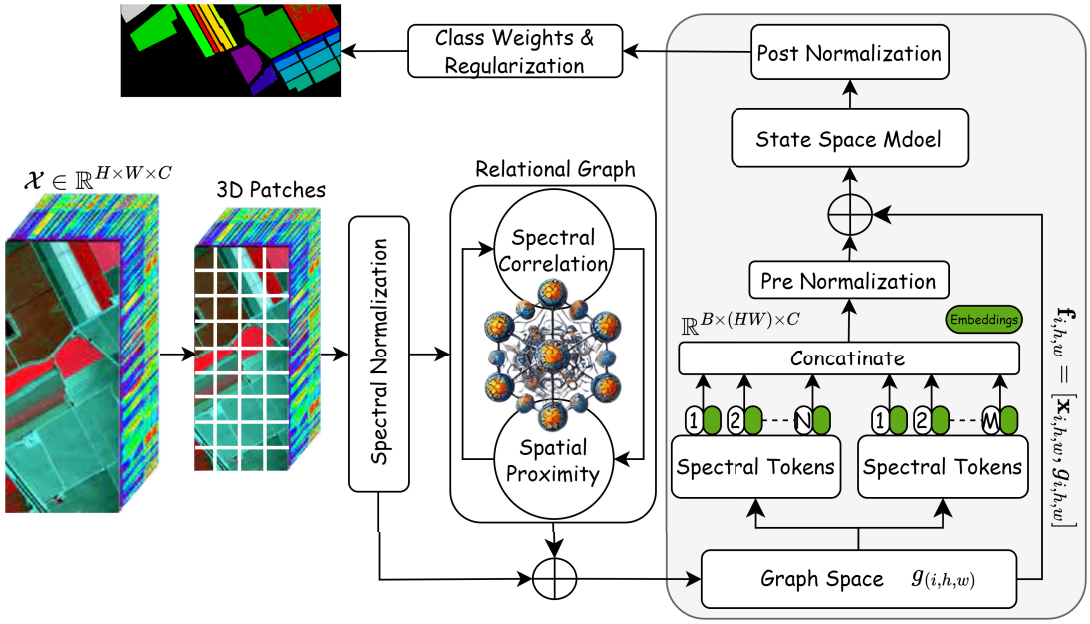


Fig. 1. RGM pipeline begins with an HSI cube, from which patches are extracted, and each spectral band is normalized. An RG is then constructed using spatial proximity and spectral similarity, forming a relational graph with locally constrained neighborhoods. Graph statistics are fused with the spectra to produce enriched per-pixel features. These features pass through a dual-branch tokenizer, consisting of a spatial and a spectral branch, yielding an embedding tensor $\mathbf{E} \in \mathbb{R}^{N_b\times L\times 2D}$. The embeddings are subsequently processed by an SSSM with an input-conditioned step Δ_t and modulation $\mathbf{B}_t$, producing the final representation $\mathbf{h}_L$. A regularized classifier head then outputs the posterior probabilities $\hat{\mathbf{y}}$.

B. Per-Band Normalization

Following patch extraction, per-band normalization is applied to reduce interband scale differences and stabilize training [64], [65]. To prevent information leakage under the disjoint protocol, the statistics are computed using training samples only. For each spectral channel $c \in \{1,\ldots,C\}$, the empirical mean μ_c and standard deviation σ_c are estimated from the training set and then applied to normalize all splits

$$\hat{x}_{i,h,w,c} = \frac{x_{i,h,w,c} - \mu_c}{\sigma_c + \epsilon}, \tag{1}$$

where ϵ is a small constant (e.g., 10^{-8}) added for numerical stability. This channelwise normalization reduces bias due to scale differences across bands, accelerates convergence, and improves generalization in scarce-sample regimes.

C. RG Initialization

The RG represents each pixel as a node [66]. For sample i, pixel coordinates (h, w) are mapped to a vertex $v_{i,h,w}$, yielding the node set $V = \{v_{i,h,w} \mid i \in \{1,\ldots,N\},\ h \in \{1,\ldots,H\},\ w \in \{1,\ldots,W\}\}$. Each node is linked to its spatial coordinates

$$v_{i,h,w} \xrightarrow{\text{hasCoordinate}} (h, w)\,. \tag{2}$$

At initialization, the RG contains only the node set and coordinate attributes, providing a consistent index for constructing spatial proximity edges and spectral affinity edges in subsequent steps. Importantly, the RG is deterministically constructed from the spatial layout and spectral measurements and is not learned from model parameters, supporting the physical interpretability of neighborhoods.

D. Spatial Proximity Relations

To establish edges in the RG based on spatial closeness, the Euclidean distance between two pixels with coordinates (h_1, w_1) and (h_2, w_2) is considered. The set of spatial edges is defined as

$$E_{\text{spatial}} = \left\{ \left(v_{i,h_1,w_1}, v_{i,h_2,w_2}\right) \;\middle|\; d\left((h_1,w_1),(h_2,w_2)\right) \le T_d \right\}, \tag{3}$$

where T_d is a threshold that determines the maximum allowable distance for two pixels to be considered spatial neighbors, and $d((h_1,w_1),(h_2,w_2)) = ((h_1-h_2)^2 + (w_1-w_2)^2)^{1/2}$. The proximity relation is encoded as

$$v_{i,h_1,w_1} \xrightarrow{\text{isNear}} v_{i,h_2,w_2} \quad \text{if} \quad d\left((h_1,w_1),(h_2,w_2)\right) \le T_d. \tag{4}$$

The threshold T_d acts as a bounded locality radius in pixel units, enforcing spatial continuity while preventing long-range geometric links that can violate locality assumptions. In practice, T_d corresponds to a small fixed-radius window, yielding sparse local neighborhoods and linear scaling with scene size. The process of constructing spatial proximity relations is illustrated in Fig. 2.

E. Spectral Similarity Relations

In addition to spatial proximity, the RG encodes spectral similarity to enforce material coherence. Let $\mathbf{x}_{i,h_1,w_1} \in \mathbb{R}^C$ and $\mathbf{x}_{i,h_2,w_2} \in \mathbb{R}^C$ denote the spectral vectors of two pixels. The set of spectral similarity edges is defined as

$$E_{\text{spectral}} = \left\{ \left(v_{i,h_1,w_1}, v_{i,h_2,w_2}\right) \;\middle|\; S\left(\mathbf{x}_{i,h_1,w_1}, \mathbf{x}_{i,h_2,w_2}\right) \ge T_s \right\}, \tag{5}$$

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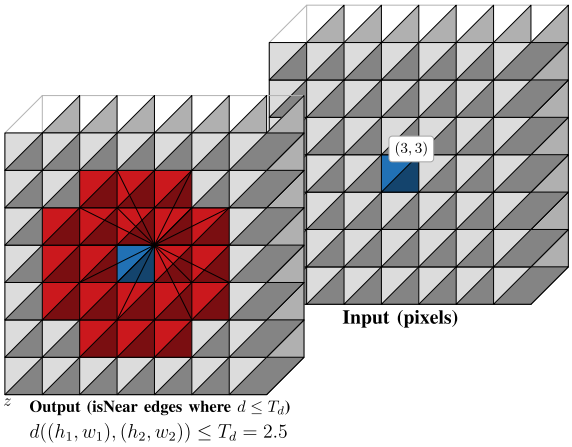


Fig. 2. **Input:** the pixel grid is represented as 3-D blocks, with the reference pixel (h_c, w_c) highlighted. **Output:** pixels within the distance threshold T_d are selected as spatial candidates and linked to (h_c, w_c) by "isNear" edges, encoding local spatial continuity in the RG.

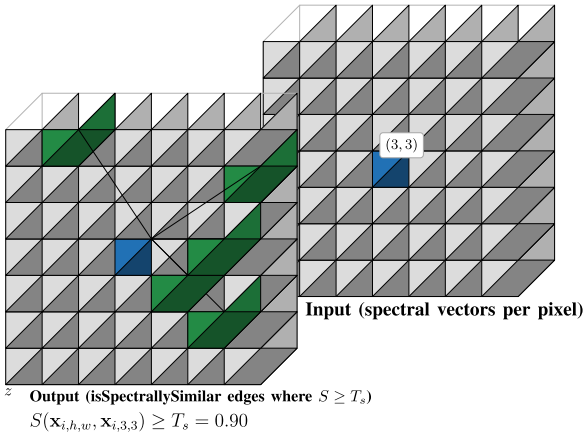


Fig. 3. **Input:** each pixel is represented by its spectral vector, and the reference pixel (h_c, w_c) is highlighted. **Output:** pixels whose cosine similarity to the reference spectrum satisfies $S \ge T_s$ are selected and linked to (h_c, w_c) via "isSpectrallySimilar" edges, enforcing spectral coherence.

where T_s is a threshold determining how aligned two spectra must be to establish a connection. The similarity function $S(\cdot)$ is based on cosine similarity

$$S\left(\mathbf{x}_i, \mathbf{x}_j\right) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\left\|\mathbf{x}_i\right\| \left\|\mathbf{x}_j\right\|}, \tag{6}$$

where $(\cdot)$ denotes the dot product and $\|\mathbf{x}\|$ is the Euclidean norm. The value of $S(\mathbf{x}_i, \mathbf{x}_j)$ lies in $[-1, 1]$, with higher values indicating stronger similarity.[1] If the similarity exceeds T_s, a spectral relation is created

$$v_{i,h_1,w_1} \xrightarrow{\text{isSpectrallySimilar}} v_{i,h_2,w_2} \quad \text{if} \quad S\left(\mathbf{x}_i, \mathbf{x}_j\right) \ge T_s. \tag{7}$$

Cosine similarity is invariant to magnitude scaling, making it less sensitive to radiometric scaling across bands and scenes. The threshold T_s is set in a high-similarity regime to favor materially coherent neighborhoods; sensitivity to this choice is evaluated in the RG sensitivity analysis. The process of building spectral similarity relations is illustrated in Fig. 3.

F. Querying the RG

Once the spatial and spectral relations are defined, the neighborhood of a reference pixel is obtained by querying nodes that satisfy both constraints:[2]

$$Q\left(v_{i,h,w}\right) = \left\{ v_{i,\hat{h},\hat{w}} \;\middle|\; \left(v_{i,h,w}, v_{i,\hat{h},\hat{w}}\right) \in E_{\text{spatial}} \cap E_{\text{spectral}} \right\}. \tag{8}$$

Equivalently, the query can be written in relational form

$$\text{Query:} \quad v_{i,h,w} \xrightarrow{\text{isNear}} ?x \quad \text{and} \quad v_{i,h,w} \xrightarrow{\text{isSpectrallySimilar}} ?x, \tag{9}$$

where $?x$ denotes candidate neighbor pixels. The intersection query is implemented as a local candidate generation followed by spectral filtering, rather than a global graph traversal. First, a bounded spatial candidate set is enumerated by direct indexing on the image grid

$$\mathcal{W}_{T_d}(h, w) = \left\{ \left(\hat{h}, \hat{w}\right) \middle| d\left((h, w), \left(\hat{h}, \hat{w}\right)\right) \le T_d \right\}, \tag{10}$$

where $|\mathcal{W}_{T_d}(h, w)| \le (2T_d + 1)^2$ is constant for fixed T_d. Second, cosine similarities are computed only for these candidates and thresholded by T_s

$$Q\left(v_{i,h,w}\right) = \left\{ \left(\hat{h}, \hat{w}\right) \in \mathcal{W}_{T_d}(h, w) \;\middle|\; S\left(\mathbf{x}_{i,h,w}, \mathbf{x}_{i,\hat{h},\hat{w}}\right) \ge T_s \right\}. \tag{11}$$

With fixed T_d, this yields linear scaling in the number of pixels $L = HW$ as $\mathcal{O}(L \cdot |\mathcal{W}_{T_d}| \cdot C) = \mathcal{O}(LC)$ and avoids materializing any dense $L \times L$ adjacency. In practice, spectra are prenormalized so that cosine similarity reduces to a dot product, and filtering and statistical aggregation are implemented with vectorized tensor operations, which are highly parallelizable. This one-hop intersection query enforces both spatial continuity and spectral coherence, producing physically grounded and interpretable relational neighborhoods that better reflect material homogeneity than purely spatial or purely spectral relations. The workflow is depicted in Fig. 4.

G. Graph-Based Feature Extraction

Graph-based features are extracted for each pixel $v_{i,h,w}$ by leveraging its queried neighborhood $Q(v_{i,h,w})$. These features encode relational evidence, allowing each pixel to be characterized by its raw spectrum and neighborhood coherence. The first feature counts the number of neighbors[3]

$$C_{i,h,w} = \left|Q\left(v_{i,h,w}\right)\right|. \tag{12}$$

If $|Q(v_{i,h,w})| = 0$, the statistics are set to zero (or a fixed default), and $C_{i,h,w} = 0$, ensuring numerical stability. Next,

[1]Note: in some formulations, $1 - S(\mathbf{x}_i, \mathbf{x}_j)$ is used as a distance measure. Here, the direct similarity form is used.

[2]This formulation avoids constructing a global adjacency matrix and instead operates on bounded local neighborhoods, ensuring scalability to large scenes

[3]These descriptors provide low-variance relational signals that are particularly effective under scarce supervision, as they summarize neighborhood consistency without introducing spatial aggregation.

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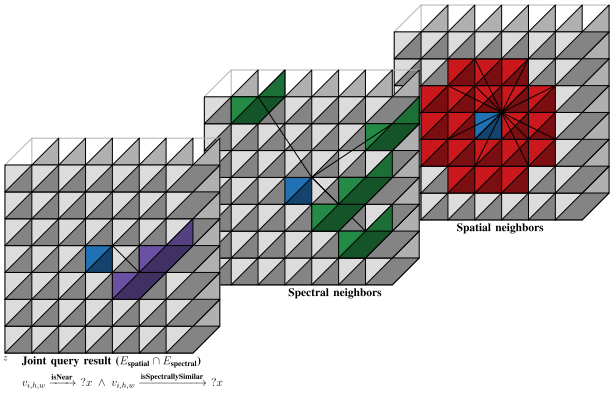


Fig. 4. Joint querying over spatial and spectral relations. **Spatial candidate:** pixels within the spatial threshold $d \le T_d$ of the reference (h_c, w_c). **Spectral affinity:** pixels whose cosine similarity to the reference spectrum satisfies $S \ge T_s$. **Intersection:** the intersection $E_{\text{spatial}} \cap E_{\text{spectral}}$ returns candidates $?x$ that are *both* spatially close and spectrally similar to v_{i,h_c,w_c}.

statistical descriptors of spectral similarity between $v_{i,h,w}$ and its neighbors are computed

$$S_{\text{avg},i,h,w} = \frac{1}{|Q\left(v_{i,h,w}\right)|} \sum_{v_{i,\hat{h},\hat{w}} \in Q\left(v_{i,h,w}\right)} S\left(v_{i,h,w}, v_{i,\hat{h},\hat{w}}\right), \tag{13}$$

$$S_{\max,i,h,w} = \max_{v_{i,\hat{h},\hat{w}} \in Q\left(v_{i,h,w}\right)} S\left(v_{i,h,w}, v_{i,\hat{h},\hat{w}}\right), \tag{14}$$

$$S_{\min,i,h,w} = \min_{v_{i,\hat{h},\hat{w}} \in Q\left(v_{i,h,w}\right)} S\left(v_{i,h,w}, v_{i,\hat{h},\hat{w}}\right), \tag{15}$$

$$S_{\text{var},i,h,w} = \frac{1}{|Q\left(v_{i,h,w}\right)|} \sum_{v_{i,\hat{h},\hat{w}} \in Q\left(v_{i,h,w}\right)} \left(S\left(v_{i,h,w}, v_{i,\hat{h},\hat{w}}\right) - S_{\text{avg},i,h,w}\right)^2. \tag{16}$$

The graph-derived feature vector is

$$F_{i,h,w} = \left[C_{i,h,w},\ S_{\text{avg},i,h,w},\ S_{\max,i,h,w},\ S_{\min,i,h,w},\ S_{\text{var},i,h,w}\right], \tag{17}$$

where $F_{i,h,w} \in \mathbb{R}^5$. Finally, the raw spectral vector $\mathbf{x}_{i,h,w} \in \mathbb{R}^C$ is concatenated with $F_{i,h,w}$ to form the enriched representation

$$\mathbf{f}_{i,h,w} = \left[\mathbf{x}_{i,h,w},\ F_{i,h,w}\right] \in \mathbb{R}^{C+5}. \tag{18}$$

The complete set of enriched pixel features for all samples is denoted by $\mathbf{F} = \{\mathbf{f}_{i,h,w}\}$.

H. Token Generation

Before the SSSM, the enriched per-pixel features are converted into a sequence of tokens. Tokenization bridges the grid-based representation and sequential modeling by disentangling local spatial aggregation from bandwise spectral mixing, preserving a one-to-one pixel-token correspondence, and producing fixed-width embeddings that are numerically well-conditioned for the SSSM. Given the per-sample enriched feature map $\mathbf{U} \in \mathbb{R}^{N_b \times H \times W \times (C+5)}$, two parallel branches are used:

1) Spatial Branch (Local Context): A 3×3 convolution aggregates information from local neighborhoods to encode short-range spatial interactions

$$\mathbf{T}^{(\text{sp})} = \phi\left(\text{Conv}^{O}_{3\times 3}\left(\mathbf{U}\right)\right), \quad \mathbf{T}^{(\text{sp})} \in \mathbb{R}^{N_b \times H \times W \times O}. \tag{19}$$

2) Spectral Branch (Band Mixing): A 1×1 convolution performs channelwise mixing without spatial smoothing

$$\mathbf{T}^{(\text{spec})} = \phi\left(\text{Conv}^{O}_{1\times 1}\left(\mathbf{U}\right)\right), \quad \mathbf{T}^{(\text{spec})} \in \mathbb{R}^{N_b \times H \times W \times O}. \tag{20}$$

The spatial dimensions are flattened to obtain $L = HW$ tokens per sample, and a light per-token projection equalizes widths

$$\widetilde{\mathbf{T}}^{(\text{sp})} = \text{Proj}\left(\text{reshape}\left(\mathbf{T}^{(\text{sp})},\ [N_b, L, O]\right)\right) \in \mathbb{R}^{N_b \times L \times O}, \tag{21}$$

$$\widetilde{\mathbf{T}}^{(\text{spec})} = \text{Proj}\left(\text{reshape}\left(\mathbf{T}^{(\text{spec})},\ [N_b, L, O]\right)\right) \in \mathbb{R}^{N_b \times L \times O}. \tag{22}$$

Concatenating the two token streams and applying LayerNorm yields stable inputs for SSSM

$$\mathbf{E} = \text{LayerNorm}\left(\left[\widetilde{\mathbf{T}}^{(\text{sp})}, \widetilde{\mathbf{T}}^{(\text{spec})}\right]\right) \in \mathbb{R}^{N_b \times L \times 2O}, \tag{23}$$

where $[\cdot,\cdot]$ denotes concatenation. Because flattening preserves the bijection $(h, w) \leftrightarrow t$, the pixel-level correspondence of the resulting classification maps is retained. This dual-branch design ensures that each token carries localized spatial context and fine-grained spectral cues that the SSSM exploits for long-range modeling.

I. State-Space Sequence Model

Let $\mathbf{E} = (\mathbf{e}_1, \ldots, \mathbf{e}_L)$ denote the token sequence per sample, with $\mathbf{e}_t \in \mathbb{R}^{2O}$. Sequence dynamics are modeled with a selective (discrete-time) state-space layer. Introduce a latent state $\mathbf{s}_t \in \mathbb{R}^{d_s}$, and at each step, input-conditioned parameters $(\mathbf{B}_t, \Delta_t)$ update this state

$$\Delta_t = \text{softplus}\left(W_\Delta \mathbf{e}_t + \mathbf{b}_\Delta\right) \in \mathbb{R}^{d_s}_{+}, \tag{24}$$

$$\mathbf{B}_t = \text{diag}\left(\sigma\left(W_B \mathbf{e}_t + \mathbf{b}_B\right)\right) \in \mathbb{R}^{d_s \times d_s}, \tag{25}$$

where σ is the logistic sigmoid, softplus ensures positive steps, and diag$(\cdot)$ converts a vector to a diagonal matrix. Starting from a time-invariant transition matrix $\mathbf{A} \in \mathbb{R}^{d_s \times d_s}$ and readout $\mathbf{C} \in \mathbb{R}^{d_s \times d_s}$, a diagonalized form is used for efficiency. The step-specific discretization is

$$\overline{\mathbf{A}}_t = \exp\left(\Delta_t \odot \boldsymbol{\lambda}\right) \in \mathbb{R}^{d_s}, \quad \overline{\mathbf{B}}_t = \left(\overline{\mathbf{A}}_t - \mathbf{1}\right) \oslash \boldsymbol{\lambda} \odot \mathbf{b}_t \in \mathbb{R}^{d_s} \tag{26}$$

where $\boldsymbol{\lambda} \in \mathbb{R}^{d_s}$ are the learned diagonal entries of $\mathbf{A}$, and $\mathbf{b}_t$ are the diagonal entries of $\mathbf{B}_t$. Here, $\odot$ and $\oslash$ denote elementwise product and elementwise division, respectively. For numerical stability, a safe version is used when $\lambda_k \approx 0$. The SSSM recurrence is

$$\mathbf{s}_t = \overline{\mathbf{A}}_t \odot \mathbf{s}_{t-1} + \overline{\mathbf{B}}_t \odot \left(W_{\text{in}} \mathbf{e}_t\right), \quad \mathbf{h}_t = W_{\text{out}}\left(\mathbf{C} \odot \mathbf{s}_t\right), \tag{27}$$

with $\mathbf{s}_0 = \mathbf{0}$ and learnable projections $W_{\text{in}} \in \mathbb{R}^{2O \times d_s}$ and $W_{\text{out}} \in \mathbb{R}^{d_s \times d_s}$. The final hidden state $\mathbf{h}_L \in \mathbb{R}^{d_s}$ summarizes global spectral–spatial context. Selectivity (via Δ_t and $\mathbf{B}_t$) adapts the memory timescale and input injection per token, enabling efficient long-range modeling while filtering noise.[4]

[4]This formulation enables adaptive, input-conditioned memory dynamics, allowing the model to selectively retain and update information across long spatial sequences.

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J. Classifier and Training Objective

The global representation $\mathbf{h}_L \in \mathbb{R}^{d_s}$ is passed through a lightweight classification head to produce class posteriors

$$\mathbf{a} = \phi(\mathbf{h}_L W_d + \mathbf{b}_d) \in \mathbb{R}^{128}, \tag{28}$$

$$\hat{\mathbf{y}} = \text{softmax}(\tilde{\mathbf{a}} W_c + \mathbf{b}_c) \in \Delta^{K-1}, \tag{29}$$

where $\tilde{\mathbf{a}} = \delta \odot \mathbf{a}$ and $\delta_j \sim \text{Bernoulli}(1-p)$ is a dropout mask with probability $p = 0.4$. The parameters are $W_d \in \mathbb{R}^{d_s \times 128}$, $\mathbf{b}_d \in \mathbb{R}^{128}$, $W_c \in \mathbb{R}^{128 \times K}$, and $\mathbf{b}_c \in \mathbb{R}^K$, where K is the number of classes and $\phi(\cdot)$ denotes a nonlinear activation (e.g., ReLU). Given a one-hot ground-truth label vector $\mathbf{y} \in \{0,1\}^K$, the objective combines cross-entropy loss with L_2 weight regularization on W_d

$$\mathcal{L} = \underbrace{-\sum_{k=1}^{K} y_k \log \hat{y}_k}_{\text{cross-entropy loss}} + \lambda \|W_d\|_F^2, \qquad \lambda = 0.01. \tag{30}$$

The cross-entropy encourages the predicted probabilities to match the ground truth, while the L_2 penalty constrains model complexity. Dropout acts as a stochastic regularizer by randomly deactivating hidden units during training, reducing coadaptation. Together, these regularization strategies reduce variance and help control capacity, which is essential in HSI classification, where labeled samples are scarce relative to input dimensionality. Thus, the classifier head translates the global spectral–spatial embedding $\mathbf{h}_L$ into reliable class predictions, while the regularized loss supports stable generalization.

K. Relational Neighborhoods

To further substantiate the physical interpretability of the proposed RG, we provide qualitative and quantitative analyses of the constructed neighborhoods. The objective is to verify that the intersection-based formulation $Q(v) = E_{\text{spatial}} \cap E_{\text{spectral}}$ captures materially coherent regions while suppressing heterogeneous mixing.

Fig. 5 presents representative examples of relational neighborhoods for pixels selected from homogeneous regions and material boundary regions. In homogeneous areas, the queried neighborhoods are spatially compact and consistent, indicating strong intraclass similarity. In contrast, for pixels located near class boundaries, the intersection criterion removes spectrally inconsistent adjacent pixels, resulting in smaller and more selective neighborhoods. This behavior demonstrates that the RG effectively enforces joint spatial–spectral consistency rather than relying on purely geometric proximity.

To complement the qualitative observations, Fig. 6 shows the distribution of graph-derived similarity statistics for homogeneous and boundary regions. Homogeneous regions exhibit higher average similarity and lower dispersion, reflecting strong material coherence. In contrast, boundary regions show reduced similarity and increased variability due to spectral transitions across classes. These results provide empirical evidence that the proposed relational neighborhoods are physically grounded and interpretable. Specifically, they adapt to local material structure, preserving homogeneous regions while reducing cross-class contamination at boundaries. This validates the design objective of enforcing material consistency through the intersection of spatial and spectral relations.

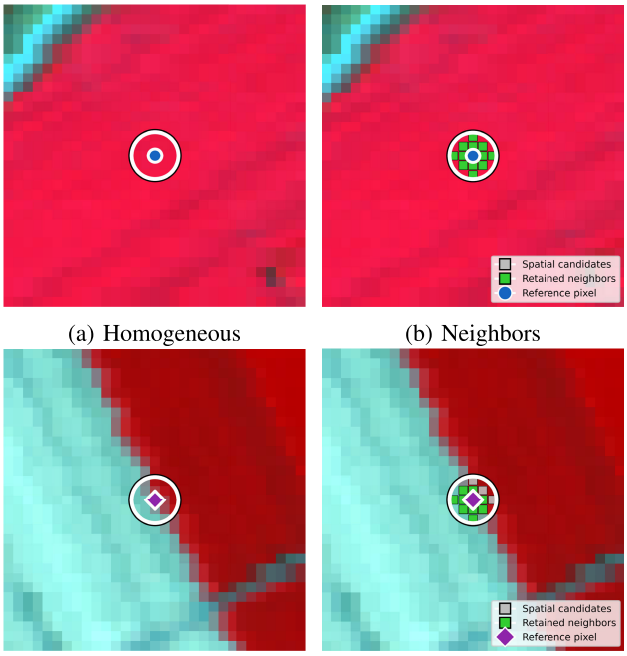


Fig. 5. Qualitative visualization of relational neighborhoods. (a) and (c) show reference pixels selected from a homogeneous region and a material boundary region, respectively. (b) and (d) illustrate the queried neighborhood $Q(v) = E_{\text{spatial}} \cap E_{\text{spectral}}$, where gray denotes spatial candidates, green denotes retained neighbors after spectral filtering, and the central marker indicates the reference pixel.

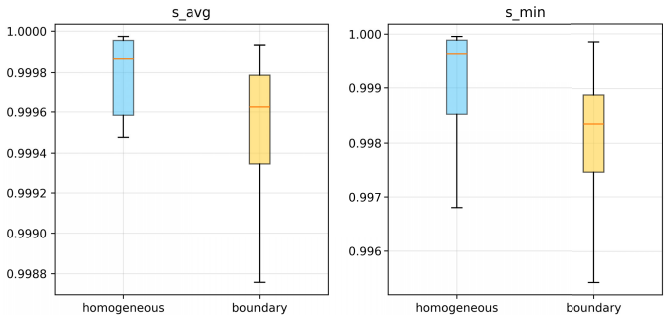


Fig. 6. Distribution of graph-derived similarity statistics in homogeneous and boundary regions. Boxplots of S_{avg} and S_{min} are shown for pixels sampled from homogeneous and boundary regions.

IV. Experimental Settings

To evaluate the effectiveness of the proposed RGM model, experiments were conducted on six widely used benchmark HSI datasets: Pavia University (PU), Salinas (SA), WHU_Hi_HongHu (HH) [67], WHU_Hi_HanChuan (HC) [67], Qingyun (QG) [68], and Pingan (PG) [68]. These datasets cover diverse sensing conditions and scene types, including urban, agricultural, and UAV-based environments. Table I summarizes the spatial size, number of classes, number of spectral channels, sensor type, and training ratio used in the scarce-label setting.

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TABLE I
SUMMARY OF THE DATASETS USED TO EVALUATE RGM

Name	Classes	Spatial Res.	Spectral Chan.	Sensor Type	Train Ratio
HH	22	940 × 475 (0.043m)	270 bands	DJI Matrice 600 Pro UAV	0.05%
HC	16	1217 × 303 (0.109m)	274 bands	Leica Aibot X6 UAV V1	0.05%
PG	10	1230 × 1000 (0.01m)	176 bands	Gaiasky mini2-VN	0.05%
QG	6	880 × 1360 (0.15m)	176 bands	Gaiasky mini2-VN	0.05%
PU	9	610 × 610 (1.3m)	103 bands	ROSIS	0.05%
SA	16	512 × 217 (3.7m)	224 bands	AVIRIS	0.05%

A. Scarce-Label Disjoint-Split Protocol

All experiments follow an extreme supervision setting in which only 0.05% of labeled pixels are used for training and 0.05% for validation, while the remaining labeled pixels are reserved for testing. To avoid spatial leakage caused by spatial autocorrelation, training, validation, and test samples are selected from nonoverlapping foreground spatial regions. This spatially disjoint evaluation protocol is used for all compared methods to ensure fairness and to better reflect realistic deployment scenarios where labeled regions are geographically separated from target areas.

B. Training Details

Unless otherwise stated, all methods are evaluated using the same input patch size ($P = 14$), identical spatially disjoint splits, and the same preprocessing pipeline. In particular, per-band normalization statistics are computed from training samples only and then applied to validation and test data to avoid information leakage. Optimization uses Adam with a cosine learning-rate schedule, a base learning rate of 1×10^{-3}, label smoothing of 0.1, a batch size 64, and 200 epochs. For the proposed RGM model, the tokenizer uses 24 output channels, the state dimension is set to $d_s = 64$, and the hidden dimension is $d_{\text{model}} = 128$. Regularization is applied using dropout ($p = 0.4$) and weight decay ($\lambda = 10^{-2}$). For baseline methods, the architectural configurations are kept consistent with their original publications and official implementations, while the training and evaluation pipeline is unified to ensure a fair comparison.

C. Evaluation Metrics

Performance is assessed using three standard metrics. Overall accuracy (OA) measures the proportion of correctly classified test pixels over the entire test set. Average accuracy (AA) is computed as the mean of per-class accuracies, reducing the influence of class imbalance. Cohen's Kappa (κ) measures agreement between predictions and ground-truth labels while accounting for chance-level agreement.

V. COMPARATIVE RESULTS AND DISCUSSION

This section evaluates HSI classification under the scarce-label, spatially disjoint protocol described in Section IV.

To contextualize the behavior of RGM and isolate architectural differences, we compare against two families of recent methods: Transformer-based models and Mamba/SSM-based models. The Transformer baselines include pyramid superpixel transformer (PF) [69], hybrid spatial–spectral transformer (HF) [70], and differential attention transformer (DF) [71]. The SSM/Mamba family includes spatial–spectral Mamba (SSM) [72], attention Mamba (AM) [56], morphological Mamba (MM) [24], graph-based Mamba (GM) [73], and policy attention-based Mamba (PM) [20]. These baselines cover complementary inductive biases, including morphology-aware, graph-guided, and hybrid attention-SSM designs. Here, we analyze the results from four perspectives: 1) quantitative performance across benchmarks, 2) qualitative consistency of classification maps, 3) accuracy-efficiency tradeoffs, and 4) the implications of spatially disjoint evaluation for scarce-label HSI classification.

A. Quantitative Performance Across Benchmarks

The most notable outcome in Table II is the consistent superiority of RGM across all six datasets in terms of κ, OA, and AA. On SA, RGM achieves $\kappa = 96.63$, OA $= 96.97$, and AA $= 97.50$, surpassing all Transformer baselines (HF: $\kappa = 92.13$, DF: $\kappa = 91.75$, PF: $\kappa = 89.99$) and the strongest Mamba competitor GM ($\kappa = 94.36$). The gain over HF and DF indicates that relational neighborhood modeling remains beneficial even in relatively homogeneous agricultural scenes.

On PU, the performance gap becomes more pronounced. PU is known for complex urban structures and spectral ambiguity among man-made materials. RGM records $\kappa = 94.50$, OA $= 95.85$, and AA $= 92.24$, compared with HF ($\kappa = 85.10$) and GM ($\kappa = 79.85$). These gains suggest that conventional self-attention and superpixel tokenization are more vulnerable under the disjoint protocol, whereas the relational descriptors used in RGM reduce boundary confusion by summarizing coherent local neighborhoods.

On HH and HC, which contain mixtures of natural and built environments, RGM again achieves clear margins: $\kappa = 94.45$ on HH and $\kappa = 95.04$ on HC, while the next-best GM reaches $\kappa = 89.21$ and $\kappa = 92.62$, respectively. These improvements correspond to meaningful error reductions in heterogeneous scenes. The PG dataset further highlights the fragility of some baselines under spatially disjoint evaluation. PF drops to AA $= 29.17$, indicating that pyramid superpixel aggregation may not generalize well when training and test regions are geographically separated. In contrast, RGM maintains AA $= 94.92$, consistent with its design goal of preserving pixel-level semantics while modeling long-range dependencies. Finally, on QG, RGM attains $\kappa = 95.36$, OA $= 96.49$, and AA $= 94.02$, outperforming HF by more than six points in κ and GM by roughly three points. Given QG's large spatial extent and spectral diversity, this result supports the scalability and robustness of the proposed design.

B. Qualitative Evaluation of Classification Maps

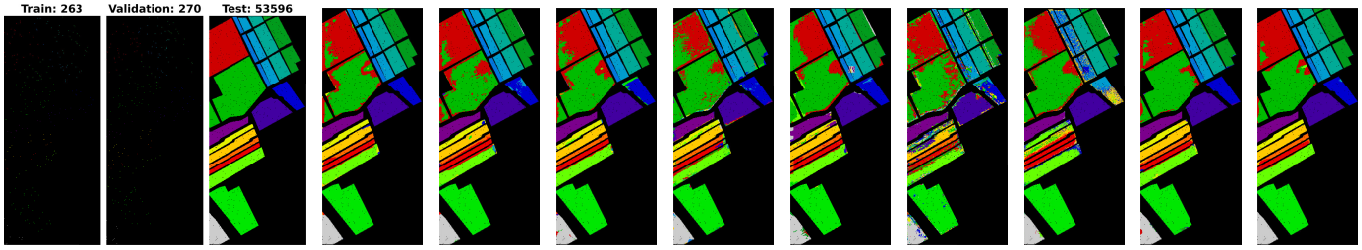
The classification maps in Figs. 7–12 visually corroborate the numerical results. Across all datasets, RGM produces

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TABLE II

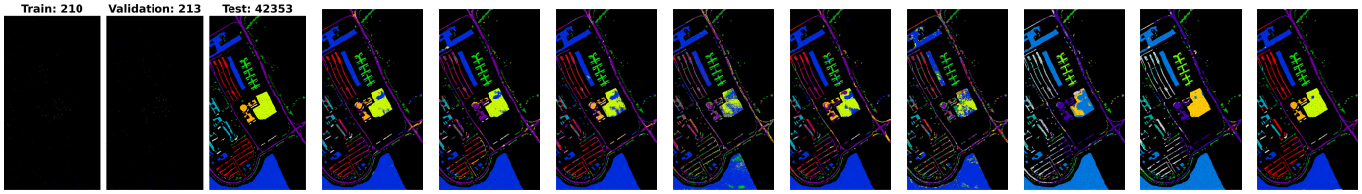
COMPARATIVE ACCURACY OF COMPETING MODELS UNDER THE SCARCE-LABEL DISJOINT-SPLIT PROTOCOL. HIGHER ACCURACIES ARE IN BOLD

Model	SA			PU			HH			HC			PG			QG		
	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA
HF	92.13	92.91	95.07	85.10	88.94	80.15	86.69	89.51	76.68	86.90	88.82	76.69	92.55	95.00	83.87	89.03	91.72	83.74
PF	89.99	90.99	92.27	80.78	85.72	70.91	82.49	86.20	68.40	85.44	87.57	76.55	69.07	79.27	29.17	87.22	59.01	42.45
DF	91.75	92.58	94.65	79.88	85.19	72.22	78.68	83.28	60.91	79.14	82.31	58.43	89.49	92.99	78.24	85.96	89.45	76.25
SSM	85.72	87.21	88.80	50.66	63.67	42.04	73.16	79.00	48.77	74.52	78.23	59.13	85.04	89.99	72.39	83.20	87.36	75.94
AM	85.61	87.08	86.64	72.25	79.55	64.22	82.48	86.22	65.97	76.79	80.38	55.76	83.98	89.37	67.13	87.49	90.58	81.37
MM	75.07	77.64	72.26	48.30	61.55	41.69	74.50	79.89	56.16	60.91	66.84	49.00	68.95	79.41	53.47	83.13	87.28	77.85
PM	74.14	76.82	64.88	60.98	71.87	49.19	86.73	89.57	68.08	87.13	88.99	76.42	92.48	94.99	82.63	85.30	88.94	78.17
GM	94.36	94.93	94.95	79.85	84.96	65.79	89.21	91.47	76.32	92.62	93.70	86.18	93.58	95.71	85.07	92.35	94.23	84.28
RGM	**96.63**	**96.97**	**97.50**	**94.50**	**95.85**	**92.24**	**94.45**	**95.60**	**92.54**	**95.04**	**95.76**	**92.24**	**95.67**	**97.07**	**94.92**	**95.36**	**96.49**	**94.02**



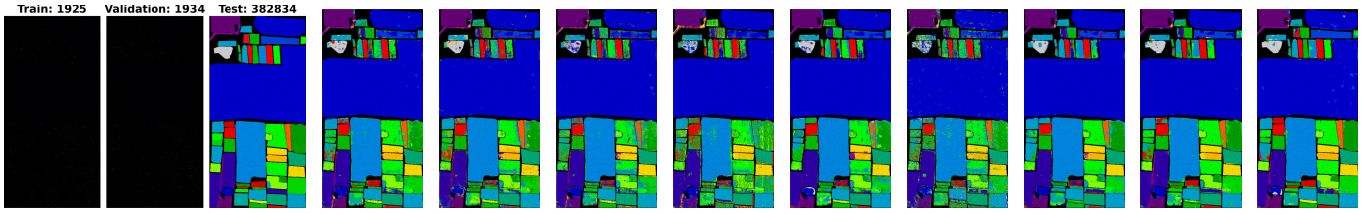
(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 7. Visual comparison on the SA dataset. (a) Ground Truths. (b) HF. (c) PF. (d) DF. (e) SSM. (f) AM. (g) MM. (h) PM. (i) GM. (j) RGM.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 8. Visual comparison on the PU dataset. (a) Ground Truths. (b) HF. (c) PF. (d) DF. (e) SSM. (f) AM. (g) MM. (h) PM. (i) GM. (j) RGM.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Visual comparison on the HH dataset. (a) Ground Truths. (b) HF. (c) PF. (d) DF. (e) SSM. (f) AM. (g) MM. (h) PM. (i) GM. (j) RGM.

cleaner maps with reduced noise and better preservation of thin or small-scale structures. On SA and PU, RGM avoids the patch-level bleeding visible in several baselines. In PU's urban areas, for example, some methods fragment roads and rooftops near class transitions, whereas RGM maintains more coherent structures by leveraging relational statistics that summarize local consistency beyond individual pixels.

In HH and HC, which contain fragmented agricultural parcels and urban elements, several baselines exhibit speckle-like artifacts and blurred transitions between neighboring regions. RGM produces sharper boundaries, particularly around narrow roads and water channels. This behavior is consistent with the pixel-aligned dual-branch tokenization, which preserves one-to-one pixel correspondence and avoids pooling-induced aliasing. On PG and QG, the differences are more pronounced. Some baselines misclassify large homogeneous regions and oversmooth fine structures, whereas RGM yields maps that remain closer to the ground truth in both global layout and local detail.

C. Why RGM Generalizes in Scarce-Label Regimes

The strong generalization of RGM under 0.05% supervision can be attributed to three complementary design choices. First, the RG intersection enforces that contextual neighborhoods are

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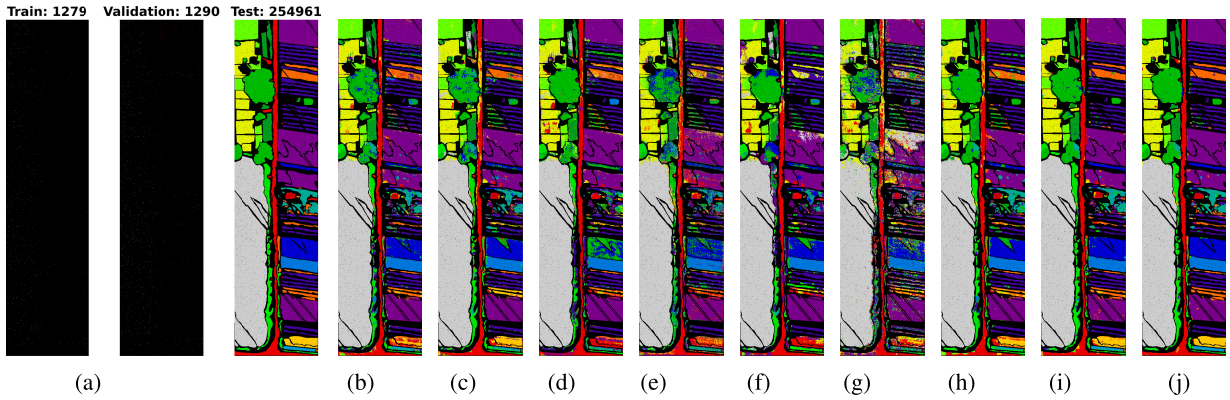


Fig. 10. Visual comparison on the HC dataset. (a) Ground Truths. (b) HF. (c) PF. (d) DF. (e) SSM. (f) AM. (g) MM. (h) PM. (i) GM. (j) RGM.

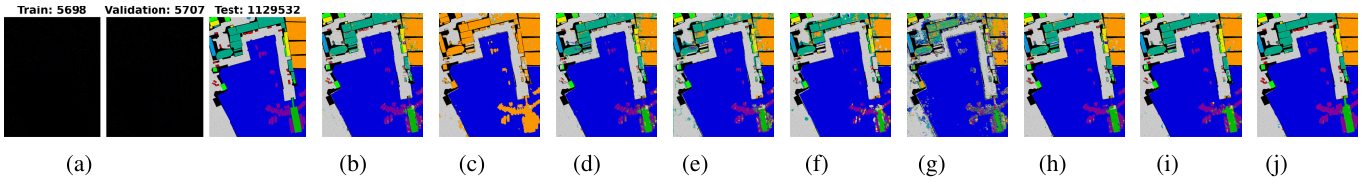


Fig. 11. Visual comparison on the PG dataset. (a) Ground Truths. (b) HF. (c) PF. (d) DF. (e) SSM. (f) AM. (g) MM. (h) PM. (i) GM. (j) RGM.

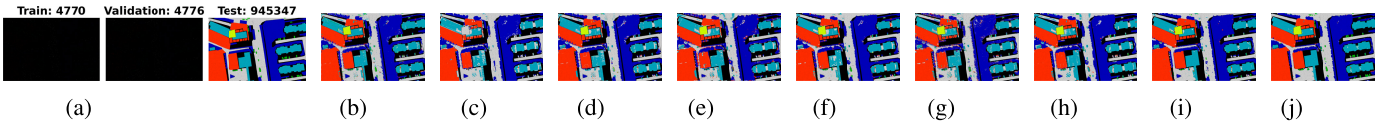


Fig. 12. Visual comparison on the QG dataset. (a) Ground Truths. (b) HF. (c) PF. (d) DF. (e) SSM. (f) AM. (g) MM. (h) PM. (i) GM. (j) RGM.

TABLE III
COMPUTATIONAL COMPARISON OF COMPETING MODELS UNDER THE SCARCE-LABEL DISJOINT-SPLIT PROTOCOL

Model	SA			PU			HH			HC			PG			QG		
	Train(s)	Test(s)	Para	Train(s)	Test(s)	Para	Train(s)	Test(s)	Para	Train(s)	Test(s)	Para	Train(s)	Test(s)	Para	Train(s)	Test(s)	Para
HF	6.44	7.21	839376	6.85	5.91	837577	39.47	171.99	840918	21.06	113.77	839376	45.98	178.60	837834	32.88	524.53	836806
PF	37.78	168.81	33778388	30.20	133.75	33777485	180.21	1210.04	33779162	138.71	804.66	33778388	1140.44	3553.98	33777614	1164.46	2956.44	33777098
DF	6.42	7.51	150256	8.92	6.56	150025	30.51	179.06	150454	16.38	116.31	150256	53.87	230.69	150058	23.12	248.56	149926
SSM	11.33	24.88	44816	13.37	21.13	43913	55.12	342.11	45590	136.54	225.92	44816	321.27	606.77	44042	166.77	534.42	43526
AM	16.68	25.57	49104	67.85	43.28	48201	113.13	359.88	49878	143.44	201.91	49104	275.86	537.15	48330	209.85	446.38	47814
MM	25.80	29.37	66752	11.25	23.29	65849	164.07	419.02	67526	156.48	223.82	66752	390.33	606.82	65978	681.41	1052.41	65462
PM	8.97	13.65	63376	9.93	10.50	62473	83.59	241.82	64150	79.78	117.29	63376	199.77	286.55	62602	108.20	593.75	62086
GM	19.80	31.77	137105	5.59	6.62	136202	30.55	166.45	137879	31.19	82.11	137105	32.44	177.96	136331	76.74	415.71	135815
RGM	179.59	68.75	54208	78.15	55.77	53305	1410.17	495.31	54982	1132.28	332.64	54208	3408.51	1476.33	53434	2876.82	1191.86	52918

both spatially proximate and spectrally similar. Unlike pure spatial graphs, which include spectrally dissimilar pixels, or pure spectral graphs, which include distant pixels, the intersection graph ensures semantic consistency. By computing distributional statistics over these neighborhoods, RGM injects low-variance descriptors that stabilize learning under scarce labels.

Second, the pixel-aligned dual-branch tokenization provides complementary spatial and spectral views without breaking the pixel-to-token correspondence. The spatial convolution captures local texture, while the spectral convolution captures bandwise signatures. This design preserves boundaries and avoids the information loss inherent in patch pooling and superpixel tokenization. Third, the SSSM backbone captures long-range dependencies in a stable and linear-time manner. Unlike transformers, which rely on quadratic attention, or RNNs, which suffer from vanishing gradients, the SSSM maintains memory across long contexts, making it particularly suited for large hyperspectral scenes.

D. Accuracy-Efficiency Tradeoffs

Table III shows that RGM is compact in terms of learnable parameters (52k-55k), substantially smaller than PF (33.7M) and also lighter than HF and DF. Nevertheless, RGM requires higher training and inference time than most baselines. This is not contradictory, but reflects the fact that the dominant cost of RGM is data-bound rather than parameter bound.

Three factors explain this behavior. First, relational preprocessing introduces overhead because RG construction and neighborhood-statistic computation are performed explicitly from spatial and spectral data. These operations add cost, but they also provide the inductive bias that improves generalization under scarce supervision. Second, pixel-accurate sequence modeling increases runtime because RGM avoids patch pooling and instead processes sequences of length $L = H \times W$. Although the SSSM scales linearly with L, large scenes still produce long token streams. Third, the disjoint protocol

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TABLE IV

ABLATION OF RGM COMPONENTS UNDER THE SCARCE-LABEL DISJOINT SPLIT. COLUMNS C1–C5 INDICATE WHETHER A COMPONENT IS ENABLED (✓) OR REMOVED (✗). C1: RELATIONAL INTERSECTION ($E_{\text{SPATIAL}} \cap E_{\text{SPECTRAL}}$); C2: NEIGHBORHOOD STATISTICS (COUNT/AVG/MAX/MIN/VAR); C3: DUAL-BRANCH TOKENIZATION (3×3 SPATIAL $+ 1 \times 1$ SPECTRAL); C4: SSSM BACKBONE; C5: REGULARIZED HEAD (DROPOUT/L2)

Variant	Components					SA			PU			HH			HC			PG			QG		
	C1	C2	C3	C4	C5	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA
Complete (RGM)	✓	✓	✓	✓	✓	96.63	96.97	97.50	94.50	95.85	92.24	94.45	95.60	92.54	95.04	95.76	92.24	95.67	97.07	94.92	95.36	96.49	94.02
(a) w/o C1: *no relational intersection* only spatial OR only spectral edges	✗	✓	✓	✓	✓	93.43	93.97	93.30	85.00	86.65	82.24	88.95	90.40	86.04	90.24	91.26	86.44	85.17	87.07	80.92	87.36	88.79	84.52
(b) w/o C2: *no neighborhood statistics* remove $[C, S_{\text{avg}}, S_{\max}, S_{\min}, S_{\text{var}}]$	✓	✗	✓	✓	✓	94.83	95.27	95.10	91.00	92.55	88.04	91.85	93.10	89.24	92.64	93.56	89.24	91.87	93.57	89.92	92.36	93.69	90.22
(c) w/o C3: *no dual-branch tokenization* single branch (either 1×1 *or* 3×3)	✓	✓	✗	✓	✓	94.23	94.67	94.40	89.90	91.45	86.54	91.05	92.40	88.14	91.94	92.76	88.24	90.47	92.17	88.12	91.26	92.59	88.72
(d) w/o C4: *no SSSM backbone* substitute with width-matched MLP	✓	✓	✓	✗	✓	89.63	90.17	89.70	83.00	84.85	79.74	84.95	86.60	81.74	86.54	87.56	82.64	83.17	85.07	79.92	85.36	86.89	82.52
(e) w/o C5: *no regularized head* remove dropout/L2	✓	✓	✓	✓	✗	95.73	96.17	96.30	92.90	94.35	90.24	93.25	94.50	90.94	93.94	94.76	90.74	93.97	95.47	92.72	93.96	95.19	92.12

reserves nearly all labeled pixels for testing, so inference is effectively performed over almost the entire labeled scene.

Methods that reduce token count more aggressively may run faster, but often at the cost of substantial accuracy degradation under disjoint evaluation, as seen for PF on PG. RGM therefore trades additional runtime for improved boundary fidelity, stronger robustness, and consistent gains across datasets. Importantly, this overhead is principled and implementation-dependent rather than a consequence of model size. In practice, runtime can be reduced through graph caching, overlap-aware tiling, vectorized similarity computation, and kernel fusion without changing the model formulation.

VI. ABLATION STUDY

A. Component-Level Ablation

To quantify the contribution of each major component, we perform ablations from the complete RGM configuration while keeping all other settings fixed, including patch extraction, optimizer, and learning-rate schedule, train-only per-band normalization, number of epochs, and batch size.

Table IV reveals a clear hierarchy among the proposed components under the scarce-label disjoint protocol. The largest degradation occurs when the SSSM backbone (C4) is removed. Replacing the SSSM with a width-matched MLP reduces κ by approximately 7–12 points across datasets (e.g., SA: 96.63 → 89.63, PU: 94.50 → 83.00), together with the largest drops in AA. This indicates that stable long-range sequence modeling over pixel-aligned tokens is particularly important when supervision is extremely limited and train/test regions are spatially separated.

The second most influential factor is relational neighborhood correctness (C1). Removing the intersection $E_{\text{spatial}} \cap E_{\text{spectral}}$ in favor of a single relation lowers κ by roughly 3–10 points, with the largest losses on heterogeneous scenes such as PU, PG, and QG. The corresponding AA drops further indicate that intersected neighborhoods reduce off-class neighbor contamination and provide lower variance contextual evidence.

Removing the dual-branch tokenizer (C3) yields moderate but consistent degradation, typically around 2–5 points in κ, confirming that the spatial (3×3) and spectral (1×1) branches provide complementary cues for boundary preservation and class discrimination. Omitting the graph-derived statistics (C2) also reduces performance, especially AA, indicating that these descriptors stabilize learning and improve minority-class robustness under scarce supervision. In contrast, removing head regularization (C5) produces the smallest loss, typically below 1.5 points in κ, suggesting that dropout and L_2 regularization are beneficial but less critical than the relational and sequence-modeling components.

B. RG Sensitivity

Table V quantifies how graph design choices affect generalization under the scarce-label disjoint protocol. Using the intersection of spatial and spectral relations ($T_d = 8$, $T_s = 0.95$) consistently outperforms the union across all datasets, with typical reductions of about 1.3–1.9 points in κ and up to about 2.3 points in AA when switching to the union. This supports the hypothesis that the union admits spectrally inconsistent but spatially adjacent neighbors, and spatially distant but spectrally similar neighbors, thereby increasing context variance. The effect is most pronounced on heterogeneous scenes such as PU and PG, where boundary mixing is stronger.

Varying T_d and the cosine threshold T_s shows the expected bias-variance tradeoff. A small neighborhood ($T_d = 4$) slightly undercontextualizes pixels, whereas a larger neighborhood ($T_d = 32$) increases the chance of including off-class neighbors, particularly reducing AA on PU and PG. Similarly, a permissive threshold ($T_s = 0.85$) admits noisier neighbors, while a very strict threshold ($T_s = 0.98$) underconnects the graph and weakens the resulting relational statistics. The default choice ($T_d = 8$, $T_s = 0.95$) consistently lies near the best operating region across datasets, balancing neighborhood purity and connectivity.

C. Tokenization Sensitivity

Figure 13 compares alternative tokenization strategies within the RGM framework. The default pixel-aligned dual-branch design consistently achieves the highest accuracy across all datasets. Replacing it with a single-branch tokenizer causes a noticeable drop: the spatial-only branch reduces κ by roughly 1–1.5 points, while the spectral-only branch leads to larger losses, confirming that complementary spatial and spectral encoding is important for boundary preservation and

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TABLE V

RG SENSITIVITY ANALYSIS. THE DEFAULT CONFIGURATION ($T_d = 8$, $T_s = 0.95$) CONSISTENTLY LIES NEAR THE OPTIMAL REGION

Variant	SA			PU			HH			HC			PG			QG		
	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA
Neighborhood formulation																		
Intersect (T_d=8, T_s=0.95)	96.63	96.97	97.50	94.50	95.85	92.24	94.45	95.60	92.54	95.04	95.76	92.24	95.67	97.07	94.92	95.36	96.49	94.02
Union (spatial ∪ spectral)	95.10	95.55	95.90	92.85	94.35	90.65	92.95	94.30	90.95	93.65	94.85	91.05	93.70	95.60	92.45	93.95	95.15	92.45
Neighborhood size (k)																		
T_d=4	96.05	96.45	96.95	93.90	95.35	91.70	93.85	95.10	91.95	94.55	95.40	91.70	94.90	96.55	94.20	94.75	95.95	93.40
T_d=16	96.40	96.80	97.35	94.25	95.60	92.00	94.20	95.40	92.25	94.85	95.60	92.05	95.35	96.85	94.65	95.10	96.25	93.70
T_d=32	95.75	96.20	96.50	93.55	95.05	91.20	93.50	94.90	91.35	94.10	95.10	91.25	94.55	96.20	93.85	94.30	95.65	92.95
Spectral threshold (τ)																		
T_s=0.85	95.90	96.30	96.65	93.60	95.10	91.40	93.65	94.95	91.60	94.30	95.20	91.45	94.65	96.30	93.95	94.45	95.80	93.05
T_s=0.90	96.35	96.75	97.20	94.10	95.50	91.90	94.05	95.30	92.10	94.75	95.50	91.95	95.15	96.70	94.50	94.95	96.10	93.45
T_s=0.98	96.20	96.60	97.05	94.00	95.45	91.85	93.95	95.20	92.00	94.65	95.45	91.90	95.05	96.65	94.45	94.85	96.00	93.35

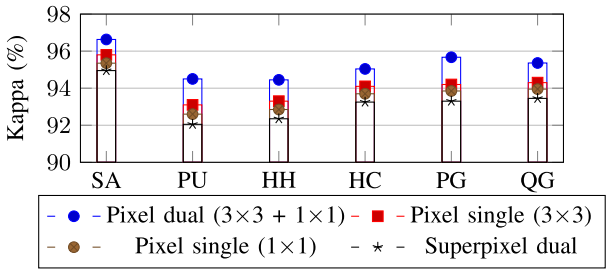


Fig. 13. Tokenization sensitivity.

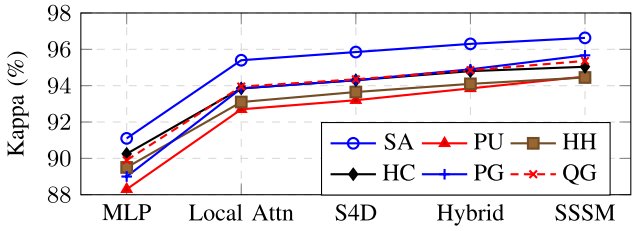


Fig. 14. Long-context modeling.

class discrimination. The superpixel-based variant yields the largest degradation, especially on PU and PG, where fine-scale structures dominate. These results indicate that preserving pixel-to-token correspondence while jointly encoding local spatial structure and spectral relationships is better suited to the disjoint, scarce-label setting than simpler alternatives.

D. Long-Context Modeling

Fig. 14 compares alternative backbones for long-range modeling within the RGM framework. Removing explicit sequence modeling and using only an MLP produces the largest degradation across all datasets, confirming that contextual memory is essential when only 0.05% of labels are available. Introducing local self-attention partially recovers performance but remains below global sequence models, especially on larger and more heterogeneous scenes such as PG and QG. The generic S4D variant improves further, indicating that stable, linear-time state-space modeling is well suited to long hyperspectral sequences. A hybrid SSM plus light attention narrows the gap but still does not match the proposed SSSM. The SSSM backbone consistently achieves the best results, supporting the benefit of input-conditioned state updates for building discriminative long-context representations.

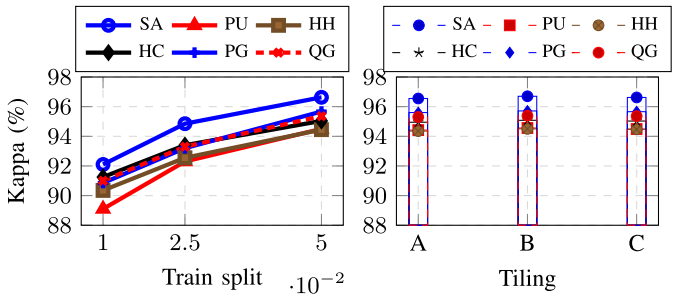


Fig. 15. Data regimes and partition sensitivity.

E. Data Regimes and Partition Sensitivity

Fig. 15 shows that RGM remains robust across both label fractions and disjoint partition geometries. In the label-fraction sweep, the model preserves strong performance even when supervision is extremely limited. At 0.01% labels, κ remains above approximately 89–92% across datasets, while at 0.05%, it reaches at least 94.5% on all scenes. The gains become smaller as more labels are added, indicating that the model can exploit its inductive biases efficiently even in very low-data settings.

The partition-sensitivity analysis further shows that these gains are not tied to a particular split. Across the three disjoint tilings, fluctuations remain within approximately 0.2–0.3 points, indicating minimal dependence on a specific spatial partition. Together, these results highlight the data efficiency and robustness of RGM in settings where annotation is scarce and train/test regions are geographically separated.

It is worth noting that, under spatially disjoint evaluation, variability arising from different train/test spatial partitions can have a greater impact on performance than stochastic effects associated with random initialization. Therefore, the present analysis focuses on partition sensitivity as a practical measure of robustness to spatial distribution shifts, which constitute the primary challenge of the adopted evaluation protocol.

VII. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

This article presented RGMamba, a relational graph-guided selective state-space model for HSI classification under

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extreme label scarcity and spatially disjoint supervision. The proposed framework produces physically grounded relational representations that improve classification under scarce-label and spatially disjoint settings. We note that interpretability is primarily associated with the relational graph construction and graph-derived feature descriptors, which explicitly encode spatial proximity, spectral affinity, and neighborhood homogeneity. In contrast, the dual-branch encoder and selective state-space model are learned neural transformations and therefore do not possess the same level of direct interpretability. In this way, RGM captures both local and long-range spectral–spatial dependencies while preserving pixel correspondence and supporting spatially coherent prediction. Experiments on six benchmark datasets showed that RGM consistently outperformed representative Transformer and Mamba baselines, with particularly strong gains under the stringent disjoint protocol using only 0.05% labeled training samples.

Despite these results, several important directions remain for future work. First, cross-scene and cross-sensor generalization remains an open challenge. Although the present study focuses on single-scene learning, practical remote-sensing applications often require robustness across sensors, seasons, and geographic regions. Incorporating domain adaptation, meta-learning, or sensor-aware normalization may improve transferability under such domain shifts. Second, extending the framework to semi-supervised and self-supervised settings is a promising direction, given the abundance of unlabeled hyperspectral data. Objectives such as contrastive pretraining, masked spectral reconstruction, or generative representation learning may further reduce dependence on manual annotation. Third, adaptive and learnable relational neighborhoods could provide greater flexibility than the current fixed construction, particularly in the presence of noise, boundary mixing, or heterogeneous local structure.

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